Validity Analysis: GLUE

Target context: Luxembourgish Public Administration Civil Servants — GLUE Assessment

Overall risk: **HIGH**

Deployment Context

Use case: In a public administration setting, a civil servant uses LLM-powered software to manage Luxembourgish citizen feedback by applying a model evaluated on topic classification, named entity recognition, intent detection, and sentiment analysis. The system automatically categorizes messages by domain and identifies key entities while classifying user intent and gauging sentiment to help detect frustrated users. These outputs enable the automated routing of requests to specific departments and the immediate prioritization of disgruntled or urgent correspondence, ensuring administrative actions are triggered accurately based on the content and tone of the message.

Target population: Civil servants in Luxembourgish government agencies who process digital correspondence and citizen inquiries written in the diverse orthographic styles of the Luxembourgish language.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ⚠	1	Serious concern	high
Input Content ⚠	1	Serious concern	high
Input Form ⚠	1	Serious concern	high
Output Ontology ⚠	1	Serious concern	high
Output Content ⚠	1	Serious concern	high
Output Form ✓	2	Significant gaps	high
Average	**1.2**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GLUE is fundamentally invalid as an evaluation benchmark for a Luxembourgish public-administration citizen-correspondence routing system. Five of six dimensions score 1 (major validity violations), with OF scoring 2 only because the surface text-in/label-out modality matches. The mismatches are total along every HIGH-priority dimension: (IO) GLUE's nine English

NLU tasks contain zero administrative-routing categories and no national-vs-communal or frontalier-vs-resident classifications [Q3, Q18, WEB-16, WEB-19]; (IC) all 409 sampled examples are English with US-political/cultural dominance, while the deployment input is trilingual lb/fr/de code-switched correspondence with non-standardized Luxembourgish orthography [WEB-11, WEB-12, DATASET-Concern 1]; (IF) GLUE assumes standardized English text input, while Luxembourgish input requires trilingual tokenization and ~21% residual orthographic noise even after best-available normalization [WEB-12]; (OO) hard single-label classification cannot represent multi-label routing with confidence scoring [Q12, DATASET-Concern 4]; (OC) annotator pools are undocumented English-speaking populations with no connection to Luxembourgish administrative register [Q26, Q36, Q73]; (OF) macro-average over nine irrelevant tasks [Q60] is uninformative as a deployment proxy. The strongest reusable value is methodological — GLUE's multi-task structure, diagnostic minimal-pair design [Q65, Q66], and human-baseline reporting [Q73, Q74] are transferable design patterns, not transferable evaluation signal.

ID	Evidence
Q3	“GLUE is a collection of NLU tasks including question answering, sentiment analysis, and textual entailment, and an associated online platform for model evaluation, comparison, and analysis.” (p.1)
Q18	“GLUE is centered on nine English sentence understanding tasks, which cover a broad range of domains, data quantities, and difficulties.” (p.3)
Q12	“All tasks are single sentence or sentence pair classification, except STS-B, which is a regression task. MNLI has three classes; all other classification tasks have two. Test sets shown in bold use labels that have never been made public in any form.” (p.2)
Q26	“The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment.” (p.3)
Q36	“The Multi-Genre Natural Language Inference Corpus (Williams et al., 2018) is a crowdsourced collection of sentence pairs with textual entailment annotations.” (p.4)
Q73	“To establish human baseline performance on the diagnostic set, we have six NLP researchers annotate 50 sentence pairs (100 entailment examples) randomly sampled from the diagnostic set.” (p.6)
Q60	“For tasks with multiple metrics (e.g., accuracy and F1), we use an unweighted average of the metrics as the score for the task when computing the overall macro-average.” (p.5)
Q65	“We construct each example by locating a sentence that can be easily made to demonstrate a target phenomenon, and editing it in two ways to produce an appropriate sentence pair.” (p.6)
Q66	“We make minimal modifications so as to maintain high lexical and structural overlap within each sentence pair and limit superficial cues.” (p.6)
Q74	“Inter-annotator agreement is high, with a Fleiss's κ of 0.73.” (p.6)
WEB-16	100 communes as of 1 September 2023 — establishes communal-routing label space requirement absent from GLUE (https://elections.public.lu/en/fusion-communes.html)
WEB-19	Loi communale 1988 defines autonomous communal competencies — establishes the legal taxonomy GLUE does not cover (https://www.sng-wofi.org/country_profiles/luxembourg.html)
WEB-11	LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic (https://arxiv.org/html/2603.10789)
WEB-12	Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' (https://arxiv.org/html/2412.09383v1)

Practical Guidance

What This Benchmark Measures

GLUE measures generalist English-language NLU competence across nine sentence-level tasks (sentiment, paraphrase, entailment, QA, acceptability, similarity) plus diagnostic linguistic-phenomenon coverage [Q3, Q15, Q18, Q135]. For the Luxembourg deployment, it provides no direct evidence of model capability — input language, content domain, output structure, and annotator perspective all diverge fundamentally. It can serve only as a structural template for designing a custom benchmark suite (multi-task scoring, diagnostic minimal pairs, human-baseline reporting), not as a deployment proxy.

Construct Depth

GLUE probes English NLU constructs reasonably deeply via its diagnostic set's four broad categories and fine-grained subcategories [Q135, Q136], but the depth is irrelevant to the deployment's constructs (administrative routing, frontalier classification, urgency detection in formal Luxembourgish register). Even the dimensions where GLUE shows methodological strength (OO metric diversity, OC inter-annotator-agreement reporting) cannot compensate for the total absence of language and domain coverage. A model achieving state-of-the-art GLUE performance could plausibly score near-zero on the actual deployment task.

What Else You Need

Comprehensive supplementation is required — GLUE cannot be remediated, only replaced or augmented. Required additions: (1) a Luxembourgish-language evaluation suite drawing on ltzGLUE [WEB-10] and LuxBorrow [WEB-11] as starting points; (2) a custom administrative-routing benchmark with État/commune competency labels, frontalier classification, and topic taxonomy elicited from CTIE/civil servants; (3) a formal-register sentiment/frustration dataset annotated by Luxembourgish civil servants to address the OC gap; (4) a code-switched lb/fr/de evaluation set covering the trilingual input distribution; (5) a multi-label-plus-confidence scoring framework matching the deployment's output requirements; (6) integration with the deployment's human-in-the-loop calibration mechanism for ongoing label-quality validation.

ID	Evidence
Q3	“GLUE is a collection of NLU tasks including question answering, sentiment analysis, and textual entailment, and an associated online platform for model evaluation, comparison, and analysis.” (p.1)
Q15	“In summary, we offer: (i) A suite of nine sentence or sentence-pair NLU tasks, built on established annotated datasets and selected to cover a diverse range of text genres, dataset sizes, and degrees of difficulty. (ii) An online evaluation platform and leaderboard, based primarily on privately-held test data. The platform is model-agnostic, and can evaluate any method capable of producing results on all nine tasks. (iii) An expert-constructed diagnostic evaluation dataset. (iv) Baseline results for several major existing approaches to sentence representation learning.” (p.2)
Q18	“GLUE is centered on nine English sentence understanding tasks, which cover a broad range of domains, data quantities, and difficulties.” (p.3)
Q135	“To make this kind of analysis feasible, we first identify four broad categories of phenomena: Lexical Semantics, Predicate-Argument Structure, Logic, and Knowledge.” (p.15)
Q136	“However, since these categories are vague, we divide each into a larger set of fine-grained subcategories.” (p.15)
WEB-10	ltzGLUE provides Luxembourgish sentiment (positive/neutral/negative) but not in administrative register (https://arxiv.org/abs/2604.17976)
WEB-11	LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic (https://arxiv.org/html/2603.10789)